

STONEWALL BANK BUOY 46050, USA

WMO: 996760

Lat: 44.677N

Lon: 124.515W

Elev: 0

StdP: 101.33

Time Zone: -8.00 (W08)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	3.4	5.0	N/A	N/A	N/A	N/A	N/A	N/A	17.4	10.1	16.5	10.3	5.9	90	0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.8	17.5	N/A	17.0	N/A	16.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	14.3	11.9	DB	1.1	18.6	2.2	1.4	-0.5	19.6	-1.7	20.4	-3.0	21.2	-4.5	22.3	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	9.5	8.9	8.8	9.5	11.5	13.4	14.3	14.2	14.0	12.8	10.9	9.4
	DBStd	2.66	1.77	1.80	1.41	1.26	1.41	1.23	1.59	1.29	1.65	1.54	1.62	2.23
	HDD10.0	182	30	38	41	23	3	0	0	0	0	1	9	37
	HDD18.3	2515	275	264	296	266	212	148	126	127	129	171	222	278
	CDD10.0	710	14	7	4	7	49	102	133	131	122	88	37	17
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.2	7.0	6.6	6.8	6.4	5.7	5.9	5.7	4.6	5.1	5.8	7.0	7.8
Precipitation	PrecAvg	2039	339	230	214	159	102	69	14	20	49	150	311	346
	PrecMax	2916	548	517	374	277	186	140	57	96	199	269	589	687
	PrecMin	1555	136	70	43	68	8	13	2	1	3	23	74	147
	PrecStd	369	120	134	81	62	47	35	13	23	44	67	133	125
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	13.7	12.5	14.3	16.1	16.8	18.0	17.9	18.2	17.3	15.0	14.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	12.8	12.5	11.6	12.4	14.4	16.0	17.4	17.3	17.4	16.1	14.2	13.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	12.2	11.8	11.1	11.5	13.9	15.6	17.0	16.8	16.8	15.5	13.6	12.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	11.7	11.1	10.7	11.1	13.5	15.2	16.6	16.3	16.4	15.0	13.0	12.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.0	2.1	2.0	1.8	1.5	1.5	1.8	2.0	2.1	1.9	2.2	2.4
		MCDBR	2.1	2.3	2.3	2.2	2.0	1.9	2.0	2.2	2.3	2.1	2.3	2.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.327	0.328	0.341	0.378	0.383	0.362	0.350	0.354	0.350	0.341	0.331	0.328	
	taud	2.524	2.543	2.525	2.379	2.402	2.484	2.540	2.537	2.540	2.556	2.537	2.498	
	Ebn at Noon	794	865	897	884	887	905	913	899	876	837	782	750	
	Edh at Noon	68	79	91	114	115	106	99	96	89	77	67	64	
All-Sky Solar Radiation	RadAvg	1.33	2.18	3.27	4.66	5.69	6.12	6.32	5.31	4.33	2.69	1.53	1.11	
	RadStd	0.18	0.22	0.35	0.32	0.43	0.38	0.47	0.48	0.32	0.26	0.19	0.15	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+17

Nomenclature: See separate page